



M1.5 Unit Overview



Exponential Functions

Students compare and contrast features of a *linear function* and an *exponential function* by creating and exploring equations, tables, and graphs. They investigate situations that grow or decay by a percentage rate per unit interval called a *growth factor*. They use the *growth factor* to write *exponential growth* or *exponential decay* functions to model and solve real situations such as those involving *compound interest*. Students fit and interpret functions that model the data between two quantities using linear or exponential functions.

📝 Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC1: Modeling with Functions, Comparing Models, CC2: Modeling with Functions, Comparing Models, Variability, CC3: Composing Functions, NS: Financial Literacy

Explore: An Epidemic

(Optional) Launch the unit with a non-routine task to investigate the question, "How does an epidemic grow?"

Investigate: Tuition Costs

(Optional) Invite students to explore tuition cost trends at post-secondary institutions. Use anytime in this course after Unit 5, Lesson 15.

Sub-Unit 1: Comparing Linear and Exponential Functions (Lessons 1–6)

- Compare and contrast linear and exponential functions.
- Create equations, tables, and graphs of linear and exponential functions.
- **Language Goal:** Describe exponential functions by their key features and statements written in function notation.

Sub-Unit 2: Exponential Growth and Decay (Lessons 7–10 + Practice Day + Quiz)

- Interpret and write exponential functions to model situations that involve percent increase or decrease.
- Use exponential functions to solve problems in context.

Sub-Unit 3: Modeling with Exponentials (Lessons 11–15 + Practice Day)

- Interpret parts of exponential expressions to make sense of compound interest rates.
- Fit a linear or exponential function to data.
- **Language Goal:** Critique the fit of a model.

Review Vocabulary

base, constant difference, constant ratio, domain, exponent, factor, initial value, interval, model, nonlinear, percent decrease, percent increase, range, rate of change, reciprocal, slope, transformation, translation, volume, y-intercept

Contextual Vocabulary

epidemic disease, toy fish, meme viral, marine biologist, colony, trade-in value, algae bloom, investment, loan, immigration, city planners

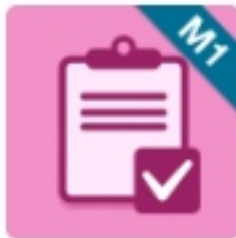
Standards

Addressing

A-CED.1	A-CED.2	A-CED.3	A-REI.11	A-REI.3	A-SSE.1b	A-SSE.2	A-SSE.1	A-SSE.1a	F-BF.1	F-BF.1a	F-BF.1b	F-BF.3	F-IF.1		
F-IF.2	F-IF.4	F-IF.5	F-IF.7	F-IF.7e	F-IF.9	F-LE.1a	F-LE.1	F-LE.1b	F-LE.1c	F-LE.3	F-LE.5	F-LE.2	N-Q.1	N-Q.2	N-Q.3
S-ID.6	S-ID.6a	SMP.1	SMP.2	SMP.3	SMP.4	SMP.5	SMP.6	SMP.7	SMP.8						

Vocabulary

- compound interest
- exponential decay
- exponential function
- exponential growth
- growth factor
- linear function
- simple interest



M1.5 Pre-Unit Check



Learn more about your students' understanding of foundational concepts and skills that will support them in the upcoming unit. This pre-unit check is intended for students to complete using a graphing calculator.

OPTIONAL This is optional and can be given in its entirety before the unit begins or spread out throughout the unit.

📧 Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions, Generalizing with Multiple Representations (Grade 6), Proportional Relationships (Grade 7), CC3: Big and Small Numbers (Grade 8)

Standards

Assessing 6.RP.3.c 7.EE.2 7.RP.3 8.EE.1 A-SSE.1.a F-BF.1.a F-IF.2 F-LE.1.b SMP.2 SMP.7 SMP.8



Explore: An Epidemic

How does an epidemic grow?

OPTIONAL This lesson is optional as the standards are addressed in other lessons within this course.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Big Idea: CC2: Comparing Models

Today's Goals

- Identify the growth pattern of an epidemic from a table.
- Recognize a graphical representation of an epidemic model.
- Language Goal:** Describe the growth pattern of an epidemic. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with the **Building Math Habits of Mind** part of this Explore to support math language acquisition for your students.

Additional Materials

scientific calculators (as needed)

Standards

Building On 8.F.2 8.F.4 8.F.5 Addressing F-LE.1 SMP.1 SMP.4 Building Toward F-IF.3 F-IF.4

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.2 ELD.PI.9-10.5 ELD.PI.9-10.6

Review Vocabulary

- rate of change
- linear
- non-linear

Instructional Routines

- Notice and Wonder
- Discussion Supports (MLR8)



Carlos's Fish

Lesson 1: Equations of Exponential Relationships

Let's make connections between exponential equations and the situations they represent.

OPTIONAL This lesson is optional as the standards are addressed in other lessons within this course.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students watch as the toy fish grows and shrinks, and they receive Responsive Feedback based on their predictions.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Comparing Models

Today's Goals

- Use equations of the form $y = a \cdot b^x$ to solve problems in context.
- Language Goal:** Interpret the parameters of an exponential equation in context. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary

mass

Additional Materials

scientific calculators (as needed)

Standards

Building On 8.EE.1 Addressing F-BF.1.a F-LE.5 SMP.1 SMP.2 Building Toward F-BF.1 F-IF.4 F-LE.1 F-LE.1.c F-LE.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.6 ELD.PI.9-10.8

Review Vocabulary

- base
- initial value

Instructional Routines

- Compare and Connect (MLR7)
- Tell a Story
- Notice and Wonder



Growing Globs

Lesson 2: Patterns of Growth

Let's identify and compare two different patterns of growth.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital manipulatives to make connections across multiple representations and visualize how patterns change over time.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Comparing Models

Today's Goals

- Identify functions that grow by constant differences as *linear functions* and functions that grow by constant ratios as *exponential functions*.
- Compare *linear* and *exponential functions* by their key features.
- Language Goal:** Explain that quantities that grow exponentially eventually exceed quantities that grow linearly. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

scientific calculators (as needed)

Standards

Building On 7.EE.3 8.F.5 F-BF.1.a Addressing F-IF.4 F-LE.1 F-LE.1.b F-LE.1.c F-LE.3 N-Q.1 SMP.1 SMP.2 SMP.3

Building Toward F-LE.1

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.5

Vocabulary

- exponential function
- growth factor
- linear function

Review Vocabulary

- constant difference
- constant ratio
- rate of change
- slope

Instructional Routines

- Tell a Story
- Notice and Wonder
- Decide and Defend
- Collect and Display (MLR2)



Going Viral

Lesson 3: Graphs and Exponential Relationships

Let's describe connections between graphs and equations and use graphs to write equations of exponential functions.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Animations and Responsive Feedback support students in making connections across multiple representations of linear and exponential functions.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions

Today's Goals

- Write exponential functions to represent graphs or situations.
- Language Goal:** Compare and contrast graphs of exponential and linear function in context. (*Writing and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

scientific calculators (as needed)

Standards

Building On 8.F.4 8.F.5 A-SSE.1 Addressing A-CED.2 A-REI.10 F-IF.4 F-LE.1 F-LE.5 F-LE.2 SMP.2 SMP.3 SMP.7

Building Toward A-REI.11 F-LE.1.c N-Q.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.6

Review Vocabulary

- initial value
- y-intercept

Instructional Routines

- Tell a Story
- Notice and Wonder



Return of the Globbs

Lesson 4: Connecting Representations of Linear and Exponential Functions

Let's make connections between different representations of linear and exponential functions.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions

Today's Goals

- Create functions, tables, and graphs of linear and exponential relationships.
- Language Goal:** Describe connections between functions, tables, graphs, and situations that represent an exponential or linear relationship. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

coloring tools (as needed); graph paper (as needed); calculators (optional)

Standards

Building On 8.F.1 8.F.5

Addressing A-CED.2 F-BF.1 F-BF.1.a F-IF.1 F-IF.4 F-IF.7 F-IF.9 F-LE.1.b F-LE.5 F-LE.2 SMP.2 SMP.3 SMP.6 SMP.7

Building Toward F-LE.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.5

Vocabulary

- exponential decay
- exponential growth

Review Vocabulary

- rate of change
- slope

Instructional Routines

- Think-Pair-Share
- Decide and Defend



Carlos and Corals

Lesson 5: Evaluating Exponential Functions

Let's evaluate exponential functions with inputs that are positive, negative, and zero.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students evaluate a function at an input and receive Responsive Feedback about where the corresponding output is located on that function's graph.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Modeling with Functions

Today's Goals

- Evaluate exponential functions with positive, zero, and negative inputs.
- Language Goal:** Describe the domain of an exponential function in context. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary

mass

Additional Materials

scientific calculators

Standards

Building On 8.EE.1 F-IF.1 Addressing F-IF.2 F-IF.5 F-IF.7e SMP.2 SMP.3 Building Toward F-BF.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.3

Review Vocabulary

- domain
- exponent
- range
- reciprocal
- volume

Instructional Routines

- Compare and Connect (MLR7)
- Collect and Display (MLR2)



Differences and Factors

Lesson 6: Changes Over Equal Intervals

Let's generalize how linear and exponential functions change over equal intervals.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC2: Comparing Models, Systems of Equations

Today's Goals

- Prove that linear functions grow by equal differences over equal intervals and that exponential functions grow by equal factors over equal intervals.
- Language Goal:** Explain how exponential functions change by a factor and linear functions change by a difference. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

chips, coins, blocks, or other counters (as needed)

Exponential Equations Activity (one per student) (optional)

Note: Consider using the *Exponential Equations* Activity after this lesson to support student understanding of using exponent properties to solve exponential equations.

Review Vocabulary

- constant difference
- constant ratio
- factor
- interval

Instructional Routines

- Think-Pair-Share

Standards

Building On 8.F.2 8.F.3 Addressing A-REI.3 F-LE.1.a F-LE.1.b SMP.2 SMP.3 SMP.7 SMP.8 Building Toward F-LE.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.5



Growing Mold

Lesson 7: Percent Increase and Decrease, Part 1

Let's explore how to model situations that change by percent increase with exponential functions.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback to understand what their responses mean in context, make connections across multiple representations, and revise their thinking.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions, CC3: Composing Functions

Today's Goals

- Model repeated percent increase with exponential functions.
- Language Goal:** Interpret the parameters of an exponential function $f(x) = a \cdot b^x$ that represents a percent increase. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On 7.EE.2 8.F.4 F-LE.1 Addressing A-SSE.1.b A-SSE.1.a F-BF.1 F-LE.1.c F-LE.2 SMP.2 SMP.7 SMP.8

Building Toward A-CED.2 F-LE.1.c F-LE.5

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6

Review Vocabulary

- percent increase

Instructional Routines

- Number Talk
- Notice and Wonder
- Decide and Defend



At a Loss

Lesson 8: Percent Increase and Decrease, Part 2

Let's make connections between different representations of exponential decay functions.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions

Today's Goals

- Calculate percent change of exponential functions from a graph.
- Write equivalent functions that represent exponential decay.
- Language Goal:** Compare and contrast equivalent equations of functions that represent exponential decay. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

graphing calculators; Activity 2 Cards (one per pair, pre-cut), coloring tools (optional)

Standards

Building On 7.EE.2 8.F.4 F-IF.1 Addressing A-SSE.1.b F-BF.1 F-LE.1.c F-LE.2 SMP.1 SMP.2 SMP.3 SMP.7 Building Toward F-BF.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.4 ELD.PI.9-10.6

Review Vocabulary

- base
- initial value
- percent decrease

Instructional Routines

- Think-Pair-Share
- Critique, Correct, Clarify (MLR3)



Lake Desmosia

Lesson 9: Solving Problems Involving Percent Growth and Decay

Let's use exponential functions to solve problems about situations that grow or decay by a percent.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and an assignable Digital Companion needed.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions

Instructional Routines

- Think-Pair-Share

Today's Goals

- Use exponential functions to solve problems in context.
- Language Goal:** Interpret the parameters a , b , and k for exponential functions of the forms $f(x) = a \cdot b^x$ and $f(x) = a \cdot b^x + k$ in context. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

CA EP&C Principle 3

Additional Materials

Activity 2 Cards (one set per pair, pre-cut)

Show What You Know Sheet

Standards

Building On F-LE.2 Addressing A-CED.2 A-SSE.1.b F-BF.1 F-BF.1.b F-BF.3 F-IF.7e F-LE.1.c F-LE.5 N-Q.1 SMP.2 SMP.7

Building Toward A-REI.11 F-BF.3 F-IF.4

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.2 ELD.PI.9-10.5



Marbleslides: Exponentials

Lesson 10: Translations of Exponential Functions

Let's practice translating exponential functions by playing a game.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students receive Responsive Feedback as they develop fluency in translating exponential functions while playing Marbleslides.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions, CC3: Composing Functions

Today's Goals

- Write functions of the form $f(x) = a \cdot b^x + k$ and $f(x) = a \cdot b^{(x-h)}$ to meet specified criteria.
- Language Goal:** Describe the effect of vertical and horizontal translations on the graph of an exponential function. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Review Vocabulary

domain, range, transformation, translation

Standards

Building On F-IF.1 F-IF.2 Addressing A-CED.2 F-BF.3 F-IF.4 F-IF.7e SMP.1 SMP.7 SMP.8 Building Toward F-IF.4 F-IF.7e

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.4 ELD.PI.9-10.5 ELD.PI.9-10.6

Instructional Routines

- Notice and Wonder
- Collect and Display (MLR2)
- Decide and Defend



M1.5 Practice Day 1

Students practice the concepts, skills, and strategies developed in Lessons 1–10. Consider using this Practice Day as preparation for the upcoming Quiz.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Modeling with Functions, Comparing Models

Materials

Option 1: Solve and Swap

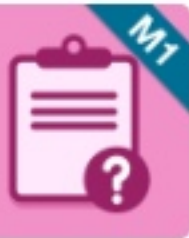
- Student Edition
- Problem Cards (one copy per seven students, pre-cut)
- graphing calculators (optional)

Option 2: Partner Practice

- Student Edition
- Problem Cards (one set per pair, pre-cut)
- graphing calculators (optional)

Standards

Assessing A-CED.2 A-SSE.1.b A-SSE.1.a F-BF.1 F-IF.2 F-LE.1 F-LE.1.c F-LE.2 SMP.1 SMP.2 SMP.7



M1.5 Sub-Unit Quiz



Learn about your students' understanding of the concepts and skills so far in this unit.
This assessment is intended for students to complete using a scientific calculator.

Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Comparing Models

Standards

Assessing

F-IF.2

F-IF.4

F-LE.1

F-LE.5

F-LE.2

SMP.2

SMP.3

SMP.4

SMP.7



Bank Accounts

Lesson 11: Introducing Simple and Compound Interest

Let's learn how to model situations involving simple and compound interest.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Repeated Challenges offer Responsive Feedback to support students in developing fluency with using the graphing calculator to solve problems involving exponential functions.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions, NS: Financial Literacy

Today's Goals

- Use a graphing calculator to solve problems involving linear and exponential functions.
- Language Goal:** Describe connections between *simple* and *compound interest*, and linear and exponential functions. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

graphing calculators (as needed); You're Invited to Explore More Sheet (optional)

Vocabulary

- compound interest
- simple interest

Instructional Routines

- Think-Pair-Share

Standards

Building On F-LE-1.a Addressing A-CED.1 A-REI.11 F-LE.1 F-LE.1.b F-LE.1.c F-LE.3 F-LE.5 SMP.2 SMP.4 SMP.5 SMP.6

Building Toward A-SSE.1.b A-SSE.1.a N-Q.1 N-Q.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.5



Payday Loan

Lesson 12: Revisiting Compound Interest

Let's analyze exponential functions that represent different compound interest scenarios.

📧 Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback to make connections across representations as they write exponential functions to represent situations involving compound interest.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Comparing Models, Modeling with Functions, NS: Financial Literacy

Today's Goals

- Write exponential functions that represent situations involving compound interest.
- **Language Goal:** Interpret parts of an exponential expression in context. (*Reading, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Materials for students using print

scientific calculators

Standards

Building On 8.EE.1 F-LE.1 Addressing A-SSE.1.a F-BF.1 N-Q.1 SMP.1 SMP.2 SMP.7 Building Toward A-SSE.1.b

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.8 ELD.PII.9-10.4

Instructional Routines

- Notice and Wonder
- Decide and Defend



Credit Card Compounding

Lesson 13: Different Compounding Intervals

Let's explore how to calculate and compare account balances with interest rates that compound at different intervals.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback to make connections across multiple representations, revise their thinking, and develop procedural fluency.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC2: Modeling with Functions, Comparing Models, NS: Financial Literacy

Today's Goals

- Calculate account balances given an initial amount, annual interest rate, compounding interval, and period of time.
- Language Goal:** Interpret parts of exponential expressions that represent annual interest rates with different compounding intervals (semi-annually, quarterly, monthly). (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary

principal

Materials for students using print

scientific calculators

Standards

Building On 7.EE.2 7.RP.3 Addressing A-SSE.1.b A-SSE.1 A-SSE.1.a F-BF.1.a N-Q.1 SMP.2 SMP.3 SMP.8

Building Toward A-SSE.1.b

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.6 ELD.PII.9-10.5

Review Vocabulary

- interval

Instructional Routines

- Collect and Display (MLR2)
- Critique, Correct, Clarify (MLR3)



Detroit's Population, Part 1

Lesson 14: Modeling Data and Goodness of Fit

Let's use functions to model the population growth of Detroit.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students explore interactive timelines of historical events and use technology to fit a model to population data.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC1: Modeling with Functions, CC2: Variability

Today's Goals

- Fit an exponential or linear function to data.
- Use linear and exponential models to make predictions.
- Language Goal:** Interpret the parameters of linear and exponential models in context. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Materials for students using print

digital lesson

Standards

Building On S-ID.6c S-ID.7 Addressing F-IF.5 F-LE.5 N-Q.3 S-ID.6 S-ID.6a SMP.1 SMP.2 SMP.3 SMP.4 Building Toward N-Q.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.3

Review Vocabulary

- domain
- model

Instructional Routines

- Notice and Wonder
- Think-Pair-Share
- Decide and Defend
- Discussion Supports (MLR8)



Detroit's Population, Part 2

Lesson 15: Modeling Exponential Data

Let's use functions to model the population of Detroit.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and an assignable Digital Companion needed.

Instruction Differentiation Beyond the Lesson

Big Ideas: CC1: Comparing Models, Modeling with Functions, CC2: Variability

Today's Goals

- Use a graphing calculator to create a linear or exponential model to fit a function to data.
- Informally assess the fit of a function to data.
- **Language Goal:** Critique the usefulness of a model based on the limitations of a context. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Activity 2 Sheet (one per student)

Additional Vocabulary

line of best fit

Standards

Building On S-ID.6 Addressing F-IF.4 F-LE.1 N-Q.3 S-ID.6 S-ID.6a SMP.4 SMP.5 Building Toward S-ID.6a

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.5

Review Vocabulary

- domain
- model

Instructional Routines

- Discussion Supports (MLR8)



M1.5 Practice Day 2

Students practice the concepts, skills, and strategies developed in Lessons 1–15. Consider using this Practice Day as preparation for the upcoming End-of-Unit Assessment.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Modeling with Functions, Comparing Models, Variability; NS: Financial Literacy

Materials

Option 1: Stations

- Student Edition
- Task cards (two sets for the whole class)
- *Combining Linear and Exponential Functions* Activity (one per student)(optional)

Option 2: Group Problems

- Student Edition
- Task cards (one set per group)
- *Combining Linear and Exponential Functions* Activity (one per student)(optional)

Note: For additional practice in combining linear and exponential functions consider using the *Combining Linear and Exponential Functions* Activity.

Standards

Addressing

A-CED.3

A-SSE.1.b

F-BF.1.a

F-BF.1.b

F-IF.2

F-IF.4

F-LE.1

F-LE.1.c

F-LE.5

F-LE.2

S-ID.6a

SMP.2

SMP.3

SMP.4



M1.5 Performance Task



Assign this summative assessment performance task at the end of the unit to evaluate students' proficiency with the concepts and skills addressed in the unit.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC2: Modeling with Functions, Comparing Models, NA: Financial Literacy

Standards

Assessing

F-BF.1.a

F-IF.5

F-LE.1

F-LE.3

F-LE.5

SMP.2

SMP.3

SMP.4



Unit Synthesis and Reflection



Here are six optional activities for students to engage in at the end of a unit to synthesize and/or reflect on their learning from the unit.

Notes:

- **Select one to two tasks** for students to complete before or after the End-Unit Assessment.
- Many tasks invite students to refer to the Student Goals and Glossary, which can be downloaded from the Unit Overview page at the beginning of each unit.

📧 Let us know what you thought about this lesson by filling out [this survey](#).

Learning Goals

- Be purposeful, strategic, and goal-oriented about learning.
- Connect different ideas from the unit and across units.
- Revisit ideas from the unit in order to make revisions and reflect on their growth.